

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 13

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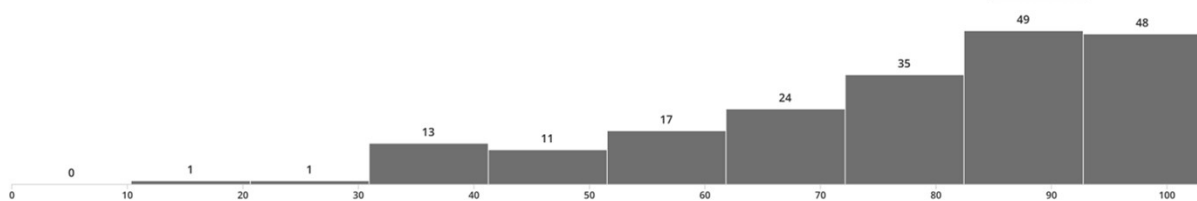
Any Reminders? Any Questions?

- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!
- Quiz on Friday on polar coordinates and beginning of triple integrals

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EXAM 1

- Average = 77, Median = 81



- Exam I – One-Page Correction (+5)

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ALVARO: Start the recording!

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More on Triple Integrals

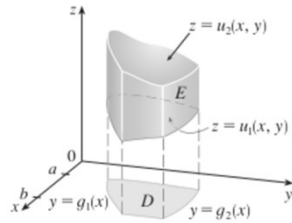
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Today – Triple Integrals

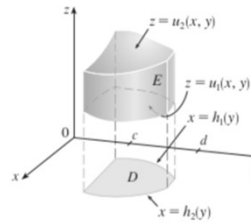
- Triple Integrals in Boxes
- Triple Integrals in Regions
- Changing the Order of Integration

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Triple Integrals in General Regions: Type I, ...



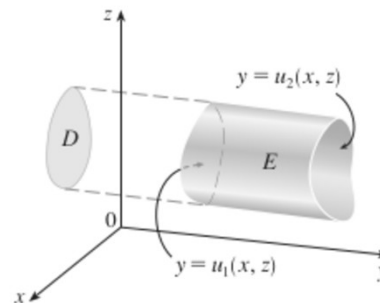
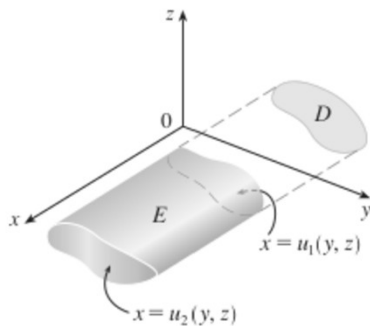
$$\iiint_E f(x, y, z) \, dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \, dx \, dy$$



$$\iiint_E f(x, y, z) \, dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \, dx \, dy$$

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Triple Integrals in General Regions: Type II and Type III



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Example: Evaluate the following triple integral

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$



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Example: Evaluate the following triple integral

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

[Extra]



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Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

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Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

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Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

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Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

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Example: Find the volume of the solid bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$.



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Example: Find the volume of the solid bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$.

[Extra]



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Questions?

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ALVARO: Start the recording!



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Triple Integrals in Cylindrical

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Today – Triple Integrals in Cylindrical Coordinates

- Cylindrical Coordinates
- Triple Integrals in Cylindrical Coordinates

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Cylindrical Coordinates

$$x = r \cos \theta \quad y = r \sin \theta \quad z = z \quad r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x} \quad z = z$$



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Example: Plot the point with cylindrical coordinates $(r, \theta, z) = (2, \pi, 3)$



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Example: Find the cylindrical coordinates of the points $(1,0,0)$, $(0,0,1)$, and $(1,1,1)$.

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Example: Describe the surface given in cylindrical coordinates by $z = r$.

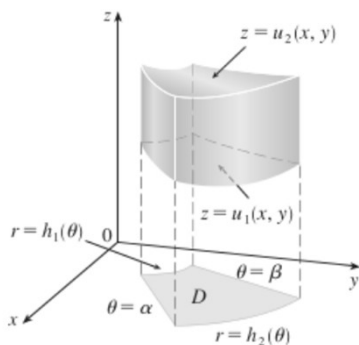
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Example: Describe the surface given in cylindrical coordinates by $z = r^2$.



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Triple Integrals in Cylindrical Coordinates (of Type I)



$$\iiint_E f(x, y, z) \, dV = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \right] dA$$

$$\iiint_E f(x, y, z) \, dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} f(r \cos \theta, r \sin \theta, z) \, r \, dz \, dr \, d\theta$$

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Example: Evaluate the value of the triple integral

$$\iiint_E x^2 dV$$

where E is the solid bounded by the surface
 $z = 4 - x^2 - y^2$ and above the xy-plane.



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Example: Evaluate the value of the triple integral

$$\iiint_E x^2 dV$$

[Extra]



$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$$



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Example: Evaluate the value of the triple integral

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) \, dz \, dy \, dx$$



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[Extra]

Example: Evaluate the value of the triple integral

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) \, dz \, dy \, dx$$

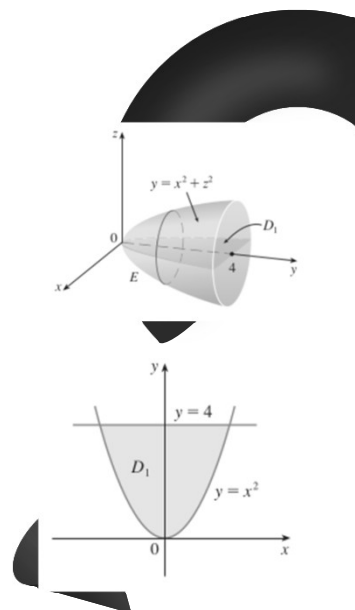


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Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + z^2} \, dV$$

where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.



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Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + z^2} \, dV$$

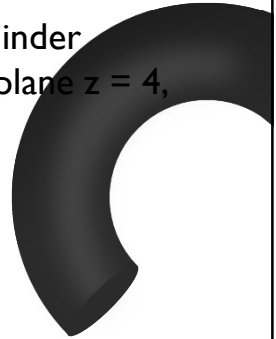
where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.

[Extra]



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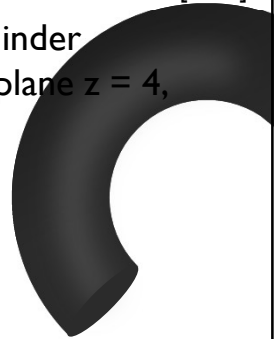
Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane $z = 4$, and above $z = 1 - x^2 - y^2$.



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[Extra]

Example: Find the volume of the solid E bounded within the cylinder $x^2 + y^2 = 1$, to the right of the xz-plane, below the plane $z = 4$, and above $z = 1 - x^2 - y^2$.



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Thank you

Until next time.

